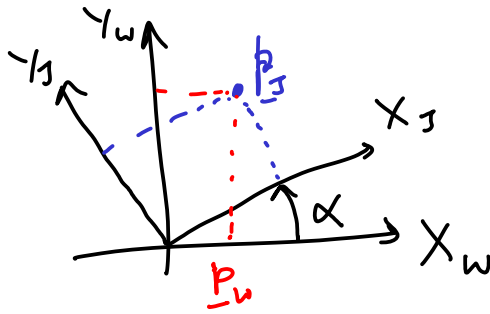


## Rotation matrix



Jetbot w.r.t.  
World is  $\alpha$

$$\underline{p}_w = {}^w R_J \underline{p}_J$$

$$\begin{bmatrix} x_w \\ y_w \end{bmatrix} = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} x_J \\ y_J \end{bmatrix}$$

① Orthonormal matrix

$${}^w R_J(\alpha)$$

$${}^w R_J^{-1} = {}^w R_J(-\alpha)$$

$${}^w R_J^T {}^w R_J = \underline{I}$$

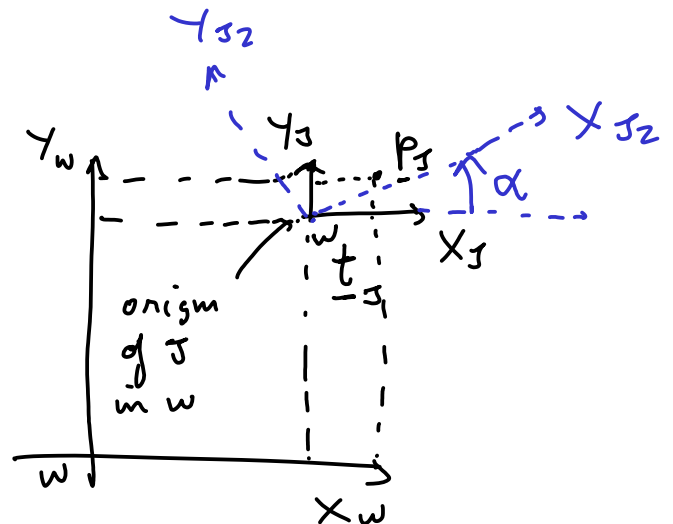
$${}^w R_J^{-1} = {}^w R_J^T = {}^J R_w$$

$$\textcircled{2} \det({}^w R_J) = 1$$

## Rotation + translation

$$\underline{p}_w = \underline{p}_J + {}^w \underline{t}_J$$

$$\underline{p}_w \xleftarrow{{}^w \underline{t}_J} \underline{p}_J \xleftarrow{R(\alpha)} \underline{p}_{J_2}$$



$$\textcircled{1} \underline{p}_J = {}^J R_{J_2}(\alpha) \underline{p}_{J_2}$$

$$\textcircled{2} \underline{p}_w = \underline{p}_J + {}^w \underline{t}_J$$

$$\underline{p}_w = {}^J R_{J_2}(\alpha) \underline{p}_{J_2} + {}^w \underline{t}_{J_2}$$

↑ Affine transformation

# Transformation matrix

$$\underline{p}_w = {}^w R_{J_2} \underline{p}_{J_2} + {}^w \underline{t}_{J_2}$$

$$\begin{bmatrix} x_w \\ y_w \end{bmatrix} = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} x_{J_2} \\ y_{J_2} \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_w \\ y_w \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{J_2} \\ y_{J_2} \\ 1 \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_w \\ y_w \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} \cos \alpha & -\sin \alpha & t_x \\ \sin \alpha & \cos \alpha & t_y \\ 0 & 0 & 1 \end{bmatrix}}_{} \begin{bmatrix} x_{J_2} \\ y_{J_2} \\ 1 \end{bmatrix}$$

homogeneous coordinates

${}^w T_{J_2}$

$$\Rightarrow \boxed{\underline{\tilde{p}}_w = {}^w T_{J_2} \underline{\tilde{p}}_{J_2}} \Leftrightarrow \begin{bmatrix} \underline{p}_w \\ 1 \end{bmatrix} = \begin{bmatrix} {}^w R_{J_2} & {}^w \underline{t}_{J_2} \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} \underline{p}_{J_2} \\ 1 \end{bmatrix}$$

Block-wise matrix multiplication works as long as the <sup>component</sup> multiplications are valid

## Properties of Transformation matrix

① Find the inverse of a transformation matrix using the fact that  $R^{-1} = R^T$

$$T = \begin{bmatrix} R_{2 \times 2} & t_{2 \times 1} \\ \underline{0}_{1 \times 2}^T & 1 \end{bmatrix}_{3 \times 3}$$

$$T^{-1} = ?$$

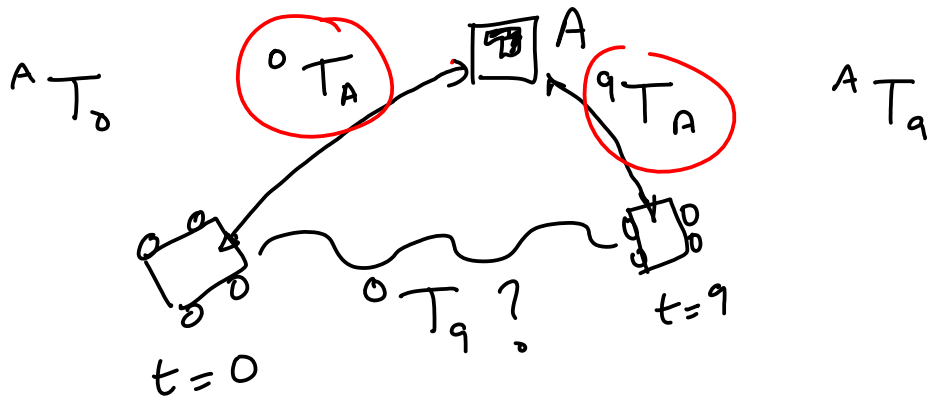
$$\begin{aligned} \underline{y} &= R \underline{x} + \underline{t} \\ \underline{x} &= R^T (\underline{y} - \underline{t}) \end{aligned}$$

$$\begin{aligned} \underline{y} &= T \underline{x} \\ \underline{x} &= T^{-1} \underline{y} \end{aligned}$$

$$\underline{x} = R^T \underline{y} - [R^T \underline{t}]_{2 \times 1}$$

$$\begin{bmatrix} \underline{x} \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} R_{2 \times 2}^T & -R_{2 \times 1}^T \underline{t} \\ \underline{0}_{1 \times 2}^T & 1 \end{bmatrix}}_{T^{-1}} \begin{bmatrix} \underline{y} \\ 1 \end{bmatrix}$$

$$T = \begin{bmatrix} R & \underline{t} \\ \underline{0}^T & 1 \end{bmatrix} \Rightarrow T^{-1} = \begin{bmatrix} R^T & -R^T \underline{t} \\ \underline{0}^T & 1 \end{bmatrix}$$



$${}^0 T_9 = ?$$

$$\underline{x}_0 = {}^0 T_9 \underline{x}_9$$

$$\underline{x}_A = {}^A T_9 \underline{x}_9$$

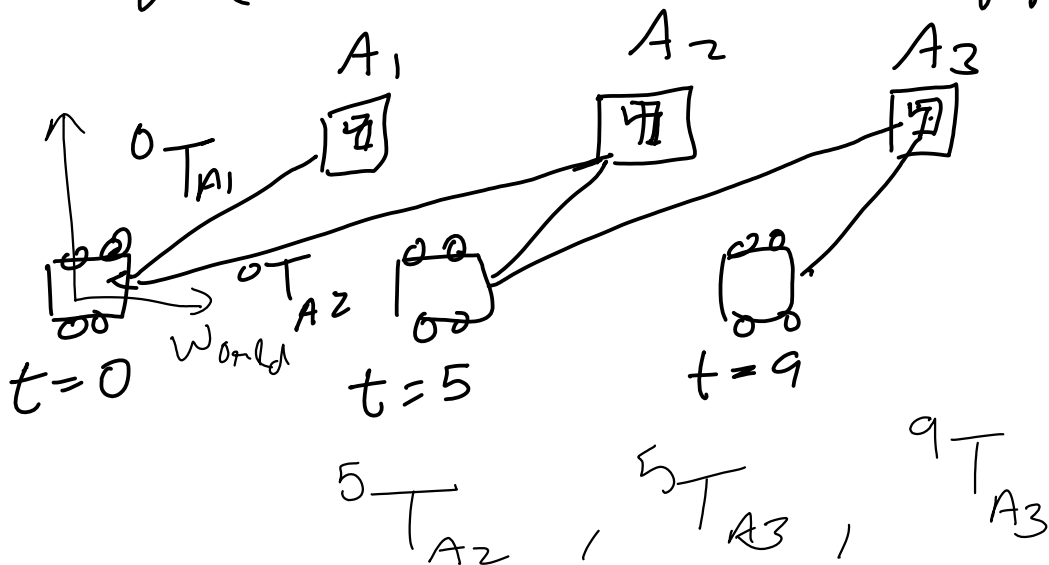
$$\underline{x}_0 = {}^0 T_A \underline{x}_A$$

$$\underline{x}_0 = \underbrace{{}^0 T_A \quad {}^A T_9}_{{}^0 T_9} \underline{x}_9$$

$$\Rightarrow \boxed{{}^0 T_9 = {}^0 T_A \quad {}^A T_9}$$

Composition of transformations

Mapping (Landmark based mapping)



$${}^0 T_{A3} = ?$$

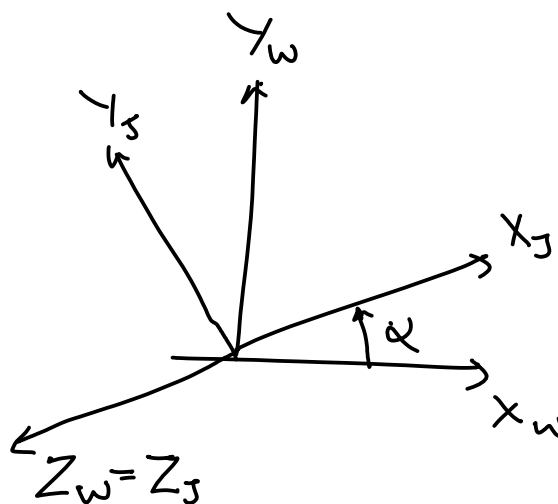
$${}^0 T_9 = ?$$

# 3D Rotations

$$\begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} = \underbrace{\begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}}_R \begin{bmatrix} x_s \\ y_s \\ z_s \end{bmatrix}$$

2D Rot

① Orthonormal  $R^T R = I$



$$R^{-1} = R^T$$

columns are unit vectors that are  $\perp$  to each other

②  $\det(R) = 1$

We can rotate in 2D and compose 2D rotations to form 3D rotations.

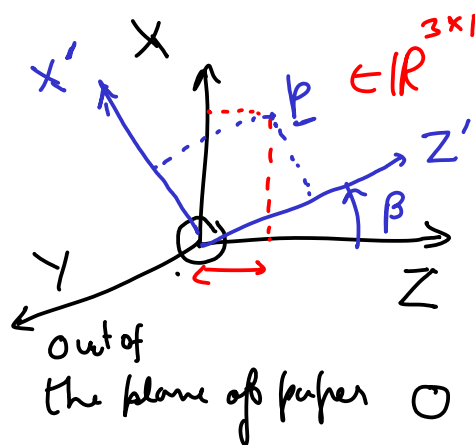
$$\rightarrow R_z(\alpha) = \begin{bmatrix} R_{2D}(\alpha) & \begin{matrix} 0 \\ 0 \end{matrix} \\ \begin{matrix} 0 & 0 \end{matrix} & 1 \end{bmatrix}_{3 \times 3} = R_{X-Y \text{ plane}}(\alpha)$$

↖ around z-axis

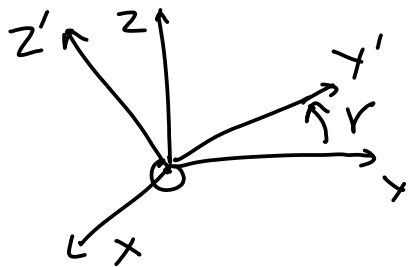
$$R_y(\beta) = \begin{bmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{bmatrix}$$

$$\begin{matrix} -1 & & \\ X & Y & \rightarrow Z \\ & & Z \times X \rightarrow Y \\ & & -1 \end{matrix}$$

$$\underset{3 \times 1}{p'} = \underbrace{R_y(\beta)}_{3 \times 3} \underset{3 \times 1}{p}$$



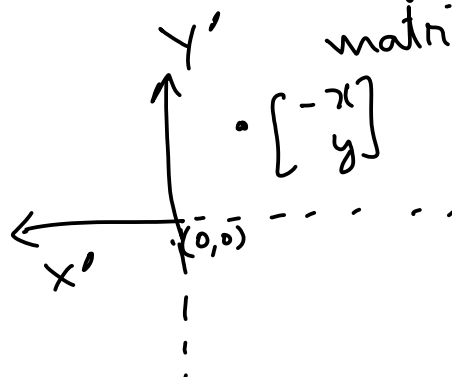
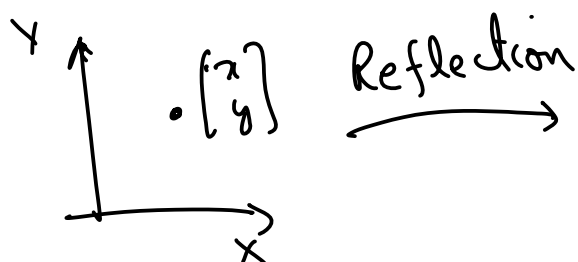
$$R_x(r) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos r & -\sin r \\ 0 & \sin r & \cos r \end{bmatrix}$$



In  $n-D$  any matrix that is Orthogonal and has determinant = 1 is a valid Rotation matrix

$$U^T U = I \Rightarrow \det(U) = +1 \text{ or } -1$$

↑  
Reflection  
matrices



$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

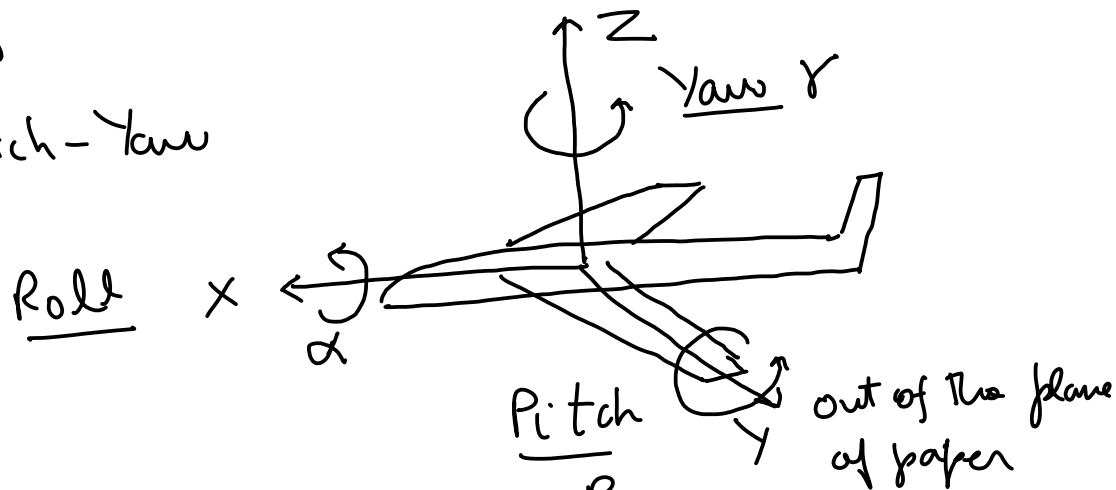
Reflection  $\neq$  Rotation

Orthogonal

$$\det(\ ) = -1$$

# Aeronautics

## Roll - pitch - Yaw



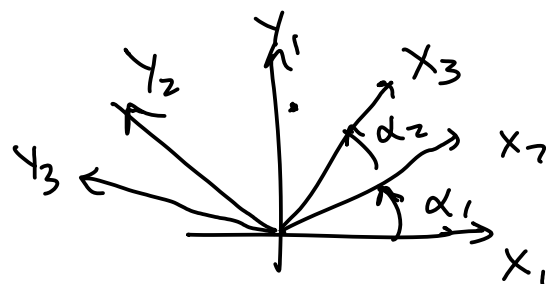
$$R(\alpha, \beta, \gamma) = R_x(\alpha) R_y(\beta) R_z(\gamma)$$

Euler angles

(12 possible

Euler angles)

Axis aligned  
Rotations



$$\underline{p}_2 = {}^2R_3 \underline{p}_3$$

$$\underline{p}_1 = {}^1R_2 \underline{p}_2$$

$$\underline{p}_1 = {}^1R_2(\alpha_1) {}^2R_3(\alpha_2)$$